

Def.

A deformation retraction of a space X onto a subspace A is a family $\{f_t: X \rightarrow X \mid t \in I\}$ satisfying $f_0 = \text{id}$, $f_1[X] = A$, $f_t|_A = \text{id}_A$ for all t , and $X \times I \rightarrow X: (x, t) \mapsto f_t(x)$ is continuous.

More generally, define

Def.

A Homotopy of a space X into a space Y is a family $\{f_t: X \rightarrow Y \mid t \in I\}$ satisfying the associated map $F: X \times I \rightarrow Y: (x, t) \mapsto f_t(x)$ is continuous.

Moreover, given two maps g_0, g_1 are called homotopic if there is a homotopy. In such a situation, denote $g_0 \simeq g_1$.

Recall that: a map $r: X \rightarrow X$ satisfying $r[X] = A$ and $r|_A = \text{id}_A$ is called retraction. Thus, we can say that the deformation retraction of a space X onto a subspace A is a homotopy from the identity map of X to a retraction of X onto A . And, algebraically, the retraction $r: X \rightarrow X$ satisfies that $r \circ r = r^2 = r$, analogous to the projection operator.

Def.

A map $f: X \rightarrow Y$ is called a homotopy equivalence if: there exists a map $g: Y \rightarrow X$ s.t. $f \circ g \simeq \text{id}_Y$ and $g \circ f \simeq \text{id}_X$.

If such a map exists, then the spaces X and Y are called homotopy equivalent, or Homotopic.

Thm. If X and Y are Homeomorphic, then X and Y are Homotopic.

Proof. Let $f: X \rightarrow Y$ be a Homeomorphism. Then,

$$f^{-1} \circ f = \text{id}_X \simeq \text{id}_X \quad \text{and} \quad f \circ f^{-1} = \text{id}_Y \simeq \text{id}_Y,$$

So proved. $\square$

Contractible

Def.

A space that homotopy equivalent to a point is called Contractible.

Thm. Let X be a Topological space. Then, TFAE:

a). X is Contractible. (Homotopy equivalent to a point)

b). $\text{id}_X: X \rightarrow X: x \mapsto x$ is null-homotopic.

c). Given $x \in X$, $C_x \simeq \text{id}_X$ where $C_x: X \rightarrow X: z \mapsto x$.

Proof.

Assume a). That is, there exist two maps

$f: X \rightarrow \{p\}$ and $g: \{p\} \rightarrow X$ such that $f \circ g \simeq \text{id}_p$ and $g \circ f \simeq \text{id}_X$.

Then, since $f: X \rightarrow \{p\}$ must be a constant mapping $x \mapsto p$,

$g \circ f(x) = g(p)$ for all $x \in X$, so $g \circ f$ is constant, so id_X is null-homotopic.

It proved b).

Assume b). Let $C: X \rightarrow X: z \mapsto x$ be a constant map such that $C \simeq \text{id}_X$, via

$H: X \times I \rightarrow X$ such that $H(z, 0) = z$ and $H(z, 1) = x$.

Let $x \in X$ be fixed. Define a path $p: [0, 1] \rightarrow X: t \mapsto H(x, t)$.

Then, $p(0) = H(x, 0) = x$ and $p(1) = H(x, 1) = x$.

Now, $F: X \times I \rightarrow X: (z, t) \mapsto p(t)$ is the Homotopy to $C \simeq C_x$. Now, by transitivity, $C \simeq \text{id}_X$ and $C \simeq C_x$ implies that $C_x \simeq \text{id}_X$. It proved c).

Assume c). Let $f: X \rightarrow \{p\}: x \mapsto p$ be a map and let $x \in X$ be given

Then, a map $g: \{p\} \rightarrow X: p \mapsto x$ satisfies

$$f \circ g = \text{id}_p \text{ and } g \circ f = C_x.$$

By c), $C_x \simeq \text{id}_X$, so $g \circ f \simeq \text{id}_X$. So proved $\square$

Corollary. Contractible space is Path-Connected.

Proof. Suppose that X is Contractible space. Let $x, y \in X$ be given,

Then, $C_x \simeq \text{id}_X \simeq C_y$, so there exists a Homotopy

$$F: X \times I \rightarrow X \text{ such that } \begin{cases} F(z, 0) = C_x(z) = x \text{ for all } z \in X \\ F(z, 1) = C_y(z) = y \text{ for all } z \in X. \end{cases}$$

Now, $p: I \rightarrow X: t \mapsto F(x, t)$ is a path from x to y . $\square$

